

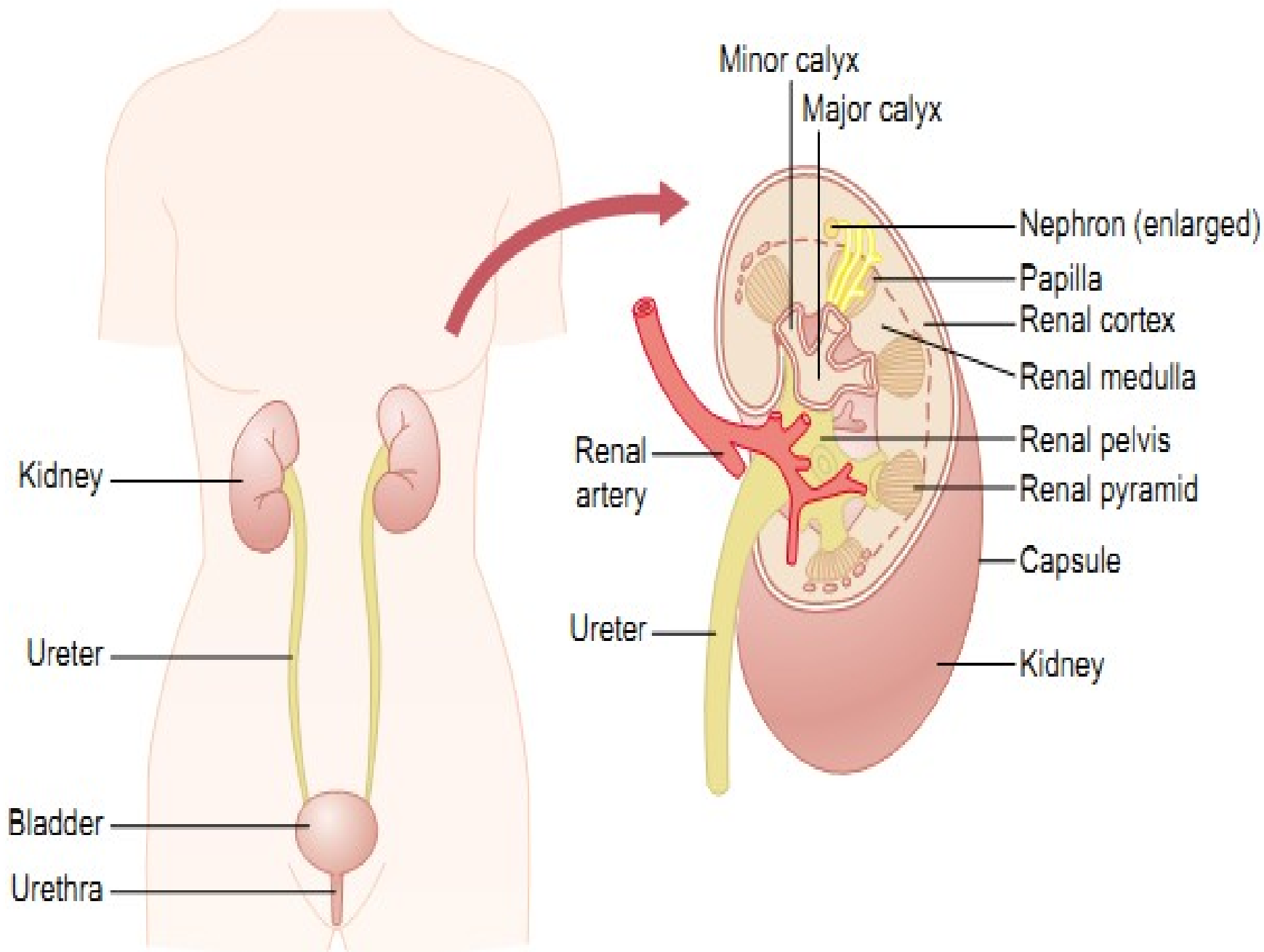
RENAL DISORDERS

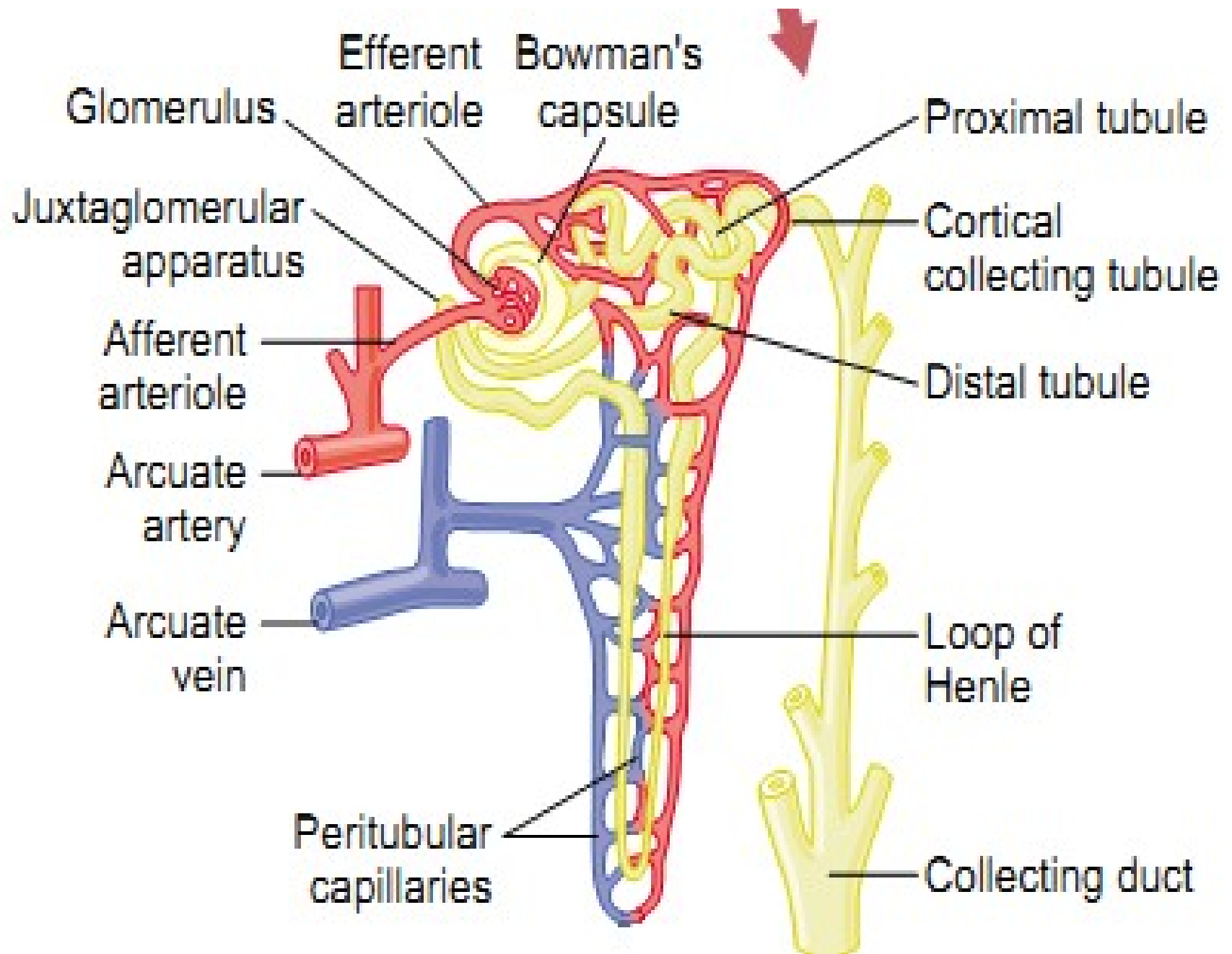
Chapter One

Introduction

2

- Functions of the kidneys in homeostasis
 1. Excretion of metabolic waste products, foreign chemicals, drugs, and hormone metabolites
 2. Regulation of water and electrolyte balances
 3. Regulation of erythrocyte production
 4. Regulation of arterial pressure
 5. Regulation of acid-base balance
 6. Regulation of 1,25–Dihydroxyvitamin D₃ (calcitriol) production
 7. Gluconeogenesis





ACUTE KIDNEY INJURY



Acute Kidney Injury

6

- **Definition:** AKI is a decrease in glomerular filtration rate (GFR) & characterized by a **sudden increase in serum creatinine** or a **reduction in urine output**
 - Generally occurring over **hours to days**, sometimes **over weeks**
 - Associated with an **accumulation of waste products** (urea and creatinine), disturbance in extracellular fluid volume and electrolyte and acid-base homeostasis.
- **Diagnostic criteria**
 - Increase in Scr by at least **0.3 mg/dL** within 48 hours or
 - Increase in Scr by at least **1.5 times** baseline within the prior 7 days or
 - Decrease in urine volume to **< 0.5 mL/kg/h for 6 hours**

Acute Kidney Injury

7

- Based on the *amount of urine output* acute renal failure may be classified as:
 - ✓ **Anuric:** if urine volume is <100 ml/day
 - ✓ **Oliguric:** if urine volume is <400 ml/day
 - ✓ **Non-oliguric:** if urine volume is ≥ 400 ml/day

Risk Factors

8

- ✍ Older age
- ✍ Higher baseline serum creatinine (SCr)
- ✍ Chronic kidney disease (CKD)
- ✍ Diabetes
- ✍ Chronic respiratory illness
- ✍ Underlying cardiovascular disease
- ✍ Prior heart surgery
- ✍ Dehydration resulting in oliguria
- ✍ Acute infection
- ✍ Renal outflow obstruction, and
- ✍ Exposure to nephrotoxins

PATHOPHYSIOLOGY

9

- ☐ Prerenal AKI
- ☐ Intrinsic AKI
- ☐ Post renal AKI

Prerenal AKI

10

- **Prerenal AKI** is the most common (55%) form of AKI
 - ↳ Represents a physiologic response to **mild to moderate renal hypoperfusion**.
- Prerenal AKI is rapidly **reversible** upon restoration of **renal blood flow** and **glomerular ultrafiltration pressure**.
- More **severe hypoperfusion** may lead to **ischemic injury** of renal parenchyma and **intrinsic renal AKI**

Prerenal AKI

11

Major causes of prerenal AKI

I. Hypovolemia

- ✓ Hemorrhage, burns, dehydration
- ✓ GI fluid loss: vomiting, surgical drainage, diarrhea
- ✓ Renal fluid loss: diuretics, osmotic diuresis (e.g., diabetes mellitus), hypoaldosteronism
- ✓ **Sequestration in extravascular space**: pancreatitis, peritonitis, trauma, burns, severe hypoalbuminemia

Prerenal AKI

12

2. **Low cardiac output:** Decrease renal blood flow

- ✓ Diseases of the myocardium, valves, and pericardium; arrhythmias; tamponade
- ✓ Other: pulmonary hypertension, massive pulmonary embolus

3. **Systemic vasodilatation:** reduction in renal perfusion pressure.

- sepsis, anaphylaxis

4. **Impairment of renal autoregulatory responses**

- ✓ NSAIDS, ACE inhibition

Intrinsic AKI

13

- Acute intrinsic AKI results from **structural damage to the kidney** itself
- Account for nearly **40%** of all AKI
 1. Renovascular obstruction (bilateral or unilateral)
 2. Disease of glomeruli or renal microvasculature
 3. Acute tubular necrosis (ATN): Ischemia, toxins
 4. Interstitial **nephritis** (inflammation within the renal parenchyma)

Post Renal AKI (Obstruction)

14

- Caused by **obstruction in the urinary tract**, leading to impaired urine flow
- Account for ~**5%** of AKI.
 - ▣ **Ureteric**: Calculi, blood clot, sloughed papillae, cancer, external compression (e.g., retroperitoneal fibrosis)
 - ▣ **Bladder neck**: Neurogenic bladder, prostatic hypertrophy, calculi, cancer, blood clot
 - ▣ **Urethra**: Stricture, congenital valve, phimosis

Signs and Symptoms of AKI

- ✓ Peripheral edema
- ✓ Weight gain
- ✓ Nausea/vomiting/diarrhea/anorexia
- ✓ Mental status changes
- ✓ Fatigue
- ✓ Shortness of breath
- ✓ Pruritus
- ✓ Volume depletion (prerenal)
- ✓ Colored/foaming urine
- ✓ Anuria alternating with polyuria (postrenal)
- ✓ Colicky abdominal pain radiating from flank to groin (postrenal)

Diagnosis

16

Laboratory Evaluation

- I. **Glomerular filtration rate(GFR):** measures of overall kidney function.
 - ✓ GFR ranges from approximately **90** to **120 mL/minute** with normal kidney function.
 - ✓ Decreased creatinine clearance in AKI
 - ✓ Direct measurement of CrCl requires collection of urine over an **extended time interval (24 hrs)** with measurement of urine volume, urine creatinine and serum creatinine levels.

Diagnosis

Urine Collection Method

$$CrCl(ml/min) = \frac{UcrV}{ScrT}$$

- ✓ Ucr = urine creatinine concentration, mg/dL
- ✓ V = volume of urine, mL
- ✓ Scr = serum creatinine concentration, mg/dL
- ✓ T = time of urine collection, minute
- ✓ (Note: time equals 1440 minutes for a 24-hour collection)

Cockcroft-Gault Equation for Adults

$$CrCl\left(\frac{ml}{min}\right) = (140 - age) \frac{BW}{Scr * 72} \times (0.85 \text{ if women})$$

Diagnosis

18

2. **Elevated serum creatinine concentration** (normal 0.6 to 1.2 mg/dL)
3. **Elevated BUN concentration** (normal 8 to 25 mg/dL)
4. **BUN: creatinine ratio**
 - elevated in prerenal AKI: $>20:1$
 - intrinsic or postrenal AKI: $<20:1$

Diagnosis

19

5. Urinalysis

- ✓ Red cell casts - vasculitis or glomerulonephritis
- ✓ Muddy brown granular casts and epithelial cell casts in a patient - acute tubular necrosis

6. Radiologic/imaging studies

- ✓ To assess urinary tract obstruction, kidney stones, renal cyst or mass
- ✓ Renal ultrasonography, plain film of the abdomen (KUB)

PREVENTION AND TREATMENT

20

Desired Outcome

- a. To prevent AKI
- b. Avoid or minimize further **renal insults** that would worsen the existing injury or delay recovery
- c. Provide supportive measures until kidney function returns

GENERAL APPROACH TO TREATMENT

21

- ✓ **Prerenal** and **postrenal** AKI can be **reversed** if the underlying problem is promptly identified and corrected, while treatment of **intrinsic AKI** failure is more supportive in nature
- ✓ Supportive therapy such as **fluid, electrolyte, and nutritional support**, renal replacement therapy (RRT), and **treatment of non-renal complications** such as **sepsis and gastrointestinal bleeding** while regeneration of the renal epithelium occurs.
- ✓ In addition, prevention of adverse drug reactions by **discontinuing nephrotoxic drugs or adjustment of drug dosages** based on the patient's renal function is desired.

Pharmacologic Therapy

22

- I. **Loop Diuretics** (furosemide, bumetanide, torsemide, and ethacrynic acid)
 - ✓ Diuretics currently have no role in preventing AKI progression or reducing mortality,
 - But they can prevent complications, such as pulmonary edema.
 - ✓ Loop diuretics are limited to instances of volume overload and edema
 - ✓ A usual starting dose of IV furosemide for the treatment of AKI is **40** mg.

Pharmacologic Therapy

23

- ✓ For edema, **IV furosemide (80 to 120 mg)** is preferred because of its potency
- ✓ Oral furosemide therapy should be avoided
- ✓ If urine output does not increase to about **1 mL/kg/hr**, the dosage can be increased to a maximum of **160 to 200 mg** of furosemide or its equivalent
- ✓ Torsemide has excellent oral BA and is unaffected by gut edema

Pharmacologic Therapy

24

- Closely monitor patient **vital signs** (weight, temperature, BP, pulse, and respirations) several times per day.
- **Diuresis** should aim for a **weight loss** of 0.5 to 1.0 kg per day.
- The patient's **volume status** should be assessed daily

Pharmacologic Therapy

25

- Other methods to improve diuresis can be initiated sequentially, such as:
 - ✍ Shortening the dosage interval;
 - ✍ Adding HCT and
 - ✍ Switching to a continuous infusion loop diuretic.
- **Thiazide diuretics**, when used as single agents, are generally not effective for fluid removal when creatinine clearance is less than **30** mL/minute.

Nonpharmacologic Therapies

26

□ Renal Replacement Therapy

✓ RRT may be necessary in patients with AKI

✍ To treat **volume overload** that is unresponsive to diuretics,

✍ To minimize the **accumulation of nitrogenous waste products**, and

✍ To correct **electrolyte and acid-base** abnormalities

RRT

27

Indications for RRT – AEIOU Criteria

A – Acid–base abnormalities (pH<7.1)	Metabolic acidosis resulting from the accumulation of organic and inorganic acids
E – Electrolyte imbalance	Hyperkalemia, hypermagnesemia
I – Intoxications	Salicylates, lithium , methanol , ethylene glycol, theophylline, phenobarbital
O – Overload(fluid)	Postoperative fluid gain/overload
U – Uremia	High catabolism of uremic toxins

Nonpharmacologic Therapies

28

□ Supportive Therapy

- ✓ Supportive therapy in AKI includes
 - ✓ Adequate nutrition,
 - ✓ Correction of electrolyte and acid-base abnormalities (particularly hyperkalemia and metabolic acidosis)
 - ✓ Fluid management, and
 - ✓ Correction of any hematologic abnormalities
- ✓ Finally, all drugs should be reviewed, and dosage adjustments made based on an estimate of the patient's glomerular filtration rate

Prevention of Acute Kidney Injury

29

- Avoidance:
 - The best preventive measure for AKI, especially in individuals at high risk, is to avoid medications that are known to precipitate AKI.
- Nephrotoxicity is a significant side effect of
 - ⊙ Aminoglycosides,
 - ⊙ Angiotensin-converting enzyme inhibitors,
 - ⊙ Angiotensin receptor antagonists,
 - ⊙ Amphotericin B,
 - ⊙ Nonsteroidal anti-inflammatory drugs,
 - ⊙ Cyclosporine, tacrolimus,
 - ⊙ Radiographic contrast agents

OUTCOME EVALUATION

30

- Maintain a state of euvolemia with good urine output (at least 1 ml/kg per hour), to return serum creatinine and BUN to baseline, and to correct electrolyte and acid-base abnormalities
- Vital signs, weight, fluid intake, urine output, BUN, creatinine, and electrolytes should be assessed daily in the unstable patient

Case study, AKI

31

- A.W. is a 71-year-old (70 KG) man who had a ST-segment elevation myocardial infarction (STEMI) 2 months ago. His ejection fraction is currently 15% (normal, 50%–60%).
- He presents today complaining of shortness of breath, dyspnea on exertion, and inability to produce much urine.
- His medical history is significant for longstanding hypertension, coronary artery disease, osteoarthritis, and recent-onset heart failure (HF) after his MI.

Case study, AKI

32

- His home medications include (all are taken orally)
 - ▣ **Furosemide** 40 mg every day,
 - ▣ **Enalapril** 5 mg daily, metoprolol XL 100 mg daily,
 - ▣ **Digoxin** 0.125 mg daily,
 - ▣ **Atorvastatin** 40 mg QD, and
 - ▣ **Naproxen sodium** 550 mg twice daily (BID),
- With the exception of naproxen, A.W. often forgets to take his medications.

Case study, AKI

33

- Physical examination reveals lower leg 3+ pitting edema, pulmonary crackles and wheezes, positive jugular venous distention, and an S₃ heart sound. His vital signs are significant for a blood pressure (BP) of 198/97 mm Hg and a weight gain of 4 kg since his last visit 2 months ago.
- Two days back, his BUN and SrCr were 23 (normal, 5–20) and 1.1 mg/dL (normal, 0.5–1.2), respectively. Current BUN and creatinine are 56 mg/dL and of 1.5 mg/dL.

Case study, AKI

34

1. What are A.W.'s risk factors for AKI?
2. What laboratory findings suggest pre-renal AKI? (dx criteria)
3. Estimate A.W.'s GFR based on CG equation
4. How should A.W.'s AKI be treated?

CHRONIC KIDNEY DISEASE

Chronic Kidney Disease

36

Definition

- a. Kidney damage for **more than 3 months**, as defined by structural or functional abnormality of the kidney, with or without decreased GFR, manifested by either pathologic abnormalities or markers of kidney damage, including abnormalities in the composition of blood or urine or abnormalities in imaging tests
- b. CKD is classified based on **C**ause, **G**FR category (G1-G5), and **A**lbuminuria category (A1-A3), abbreviated as CGA.

KDIGO GFR and albuminuria categories and prognosis of CKD by category				Persistent albuminuria		
				A1	A2	A3
				Normal to mildly increased	Moderately increased	Severely increased
				<30mg/g	30 -300 mg/g	>300 mg/g
GFR categories (mL/min/1.73m²)	G1	Normal or high	≥90			
	G2	Mildly decreases	60 – 89			
	G3a	Mild to moderate	45 – 59			
	G3b	Moderate to severe	30 – 44			
	G4	Severely decreased	15 – 29			
	G5	End-Stage Kidney Disease, ESRD	< 15			
	Low risk				High risk	
	Moderately increased risk				Very high risk	

ETIOLOGY

39

I. Susceptibility Factors

- ✓ Associated with an **increased risk**, but not proven to directly cause CKD
 -  Advanced age,
 -  Low income or education,
 -  Racial/ethnic minority status,
 -  Reduced kidney mass,
 -  Low birth weight,
 -  Family history of CKD,
 -  Systemic inflammation,
 -  Dyslipidemia
- ✓ Mostly **not modifiable**

ETIOLOGY

40

2. Initiation Factors

- ✓ Initiation factors are conditions that directly **result in kidney damage**
- ✓ **Modifiable** by pharmacologic therapy.
 -  Diabetes mellitus,
 -  Hypertension,
 -  Autoimmune diseases,
 -  Polycystic kidney disease,
 -  Systemic infections,
 -  Urinary tract infections,
 -  Urinary stones,
 -  Lower urinary tract obstructions, and
 -  Drug toxicity

ETIOLOGY

41

3. Progression Factors

- ✓ Progression risk factors are those associated with **further kidney damage**.
- ✓ The most important predictors of progressive CKD
 -  Proteinuria
 -  Elevated blood pressure
 -  Hyperglycemia
 -  Hyperlipidemia,
 -  Smoking

PATHOPHYSIOLOGY

42

- Damage to the kidney results from any of the **initiation factors** which resulting in **loss of nephron mass**.
- The resultant effect is **hypertrophy of the remaining nephrons** to **compensate for the loss of renal function and nephron mass**.
- These *adaptive* changes result in an **increase in glomerular filtration and tubular function**, both reabsorption and secretion, in the remaining nephrons
- As **time progresses**, **glomerular capillary pressure is increased**, mediated by **angiotensin II**, to maintain the **hyper filtration state of the functioning nephrons**.

PATHOPHYSIOLOGY

43

- Increased glomerular capillary pressure **expands the pores** in the glomerular basement membrane, altering the **size-selective barrier** and allowing **proteins** to be filtered through the glomerulus.
- Filtered proteins are **reabsorbed** in the **renal tubules**, which activates the tubular cells to produce **inflammatory and vasoactive cytokines** and **triggers complement activation**.
- These in turn cause **interstitial damage and scarring** within the **renal tubules**, leading to damage and **loss of more nephrons**.

CLINICAL PRESENTATION OF CKD

44

□ Symptoms

- ✓ Stages **1** and **2** CKD are generally asymptomatic.
- ✓ Stages **3** and **4** CKD may be associated with minimal symptoms.
- ✓ Typical symptoms associated with stage **5** CKD include pruritus, dysgeusia, nausea, vomiting, constipation, muscle pain, fatigue, and bleeding abnormalities.

CLINICAL PRESENTATION OF CKD

45

□ Signs

- ✓ **Cardiovascular**: worsening hypertension, edema, dyslipidemia, left ventricular hypertrophy, electrocardiographic changes and chronic heart failure
- ✓ **Genitourinary**: changes in **urine volume** and **consistency**, “foaming” of **urine** (indicative of **proteinuria**), and **sexual dysfunction**

Laboratory Tests for CKD

46

1. Increased BUN and SCr and decreased GFR
2. Increased K, P, and Mg; decreased bicarbonate (metabolic acidosis); calcium levels are generally low in earlier stages of CKD and may be elevated in stage 5 CKD, secondary to the use of calcium-containing phosphate binders.
3. Decreased albumin, if inadequate nutrition intake in advanced stages.
4. Decreased RBC count, Hgb and Hct

Laboratory Tests for CKD

47

5. Proteinurea/albuminurea: urine positive for albumin or protein
 - ✓ Normal: **<30mg/24hrs**
 - ✓ Microalbuminurea: **30-300mg/24hrs**
 - ✓ Macroalbuminurea (overt proteinurea): **>300mg/24hrs**
 - ✓ Nephrotic range proteinurea: **>3g/24hrs.**
6. Increased parathyroid hormone (PTH) level
7. Decreased vitamin D levels (stages 4 or 5 CKD).

TREATMENT OF CKD

48

□ Desired Outcomes

- ✓ The primary goal is to **slow and prevent** the progression of CKD

TREATMENT OF CKD

49

Nonpharmacologic Therapy

□ Dietary Protein Restriction:

- ✓ Shown to slow the progression of kidney disease.
- ✓ However, protein restriction must be balanced with the risk of **malnutrition** in patients with CKD.
- ✓ Patients with a GFR less than **25mL/min/1.73m²** received the most benefit from **protein restriction**; therefore, patients with a GFR above this level should not restrict protein intake

TREATMENT OF CKD

50

- Patients who have a GFR <25 mL/min / 1.73 m² who are not receiving dialysis should restrict protein intake to **0.6 g/kg per day**.
- Malnutrition is common in patients with ESRD due to:
 - ✓ Decreased appetite
 - ✓ **Hypercatabolism**
 - ✓ Nutrient losses through dialysis.
- For this reason, patients receiving dialysis should maintain protein intake of **1.2 – 1.3 g/kg per day**

TREATMENT OF CKD

51

Pharmacologic Therapy

1. Intensive Blood Glucose Control:

- ✓ Intensive insulin therapy to maintain **preprandial blood glucose** levels between **70** and **125** g/dL and postprandial blood glucose levels less than **180** g/dL

2. Optimal Blood Pressure Control:

- ✓ Stages **1-4 of CKD**: goal BP of less than **130/80** mm Hg
- ✓ Stage **5 of CKD**: receiving hemodialysis, **goal BP of <140/90** mm Hg before hemodialysis and **<130/80** after hemodialysis.

TREATMENT OF CKD

52

3. Reduction in Proteinuria

- **ACE inhibitors and ARBs:**
 - ✓ Decrease glomerular capillary pressure and volume because of their effects on angiotensin II.
 - ✓ This, in turn, reduces the amount of protein filtered through the glomerulus, independent of the reduction in blood pressure
 - ✓ Antihypertensive agents of choice for all patients with CKD.

TREATMENT OF CKD

53

4. Hyperlipidemia Treatment

- ✓ CKD patients are among the **highest-risk group** for **cardiovascular conditions** (i.e., equivalent to that of patients with known **coronary heart disease**)
- ✓ Should be treated aggressively for dyslipidemia to an LDL cholesterol goal below **100 mg/dL**

COMPLICATIONS OF CKD

54

I. Fluid and Electrolyte Abnormalities

a) Impaired Sodium and Water Homeostasis

- Sodium and water balance are primarily regulated by the kidney.
- Reductions in nephron mass-↓ GF → reabsorption of sodium and water → edema
- Treatment
 - ✓ **Nonpharmacologic:** Sodium (<2 g/day) and fluid restrictions
 - ✓ **Pharmacologic:** Loop diuretics therapy

COMPLICATIONS OF CKD

55

b) Impaired Potassium Homeostasis

- ✓ Potassium balance is primarily regulated by the kidney via the **distal tubular cells**.
- ✓ Reduction in **nephron mass** decreases tubular excretion of potassium, leading to **hyperkalemia**.
- ✓ Nonpharmacologic: **avoid abrupt increases in dietary intake of potassium, dialysis**

COMPLICATIONS OF CKD

56

b) Impaired Potassium Homeostasis

- **Pharmacologic**

- ✓ Patients who present with **cardiac abnormalities** caused by hyperkalemia should receive **calcium gluconate** or **chloride** (1 g IV) to reverse the **cardiac effects**
- ✓ Temporary measures can be employed to shift **extracellular potassium** into the **intracellular compartment** to stabilize cellular membrane effects of excessive serum potassium levels.
 - ✓ The use of **regular insulin** (5 to 10 units IV) and **dextrose** (5% to 50% IV), or **nebulized albuterol** (10 to 20 mg).

COMPLICATIONS OF CKD

57

Impaired Potassium Homeostasis...

- These measures will decrease serum potassium levels within 30 to 60 minutes after treatment, but potassium must still be removed from the body by dialysis.
- Often, **multiple dialysis sessions** are required to remove potassium that is redistributed from the intracellular space back into the serum.
- **Sodium polystyrene sulfonate** (SPS, 15 to 30 g orally), a sodium-potassium exchange resin, promotes potassium excretion from the GI tract

COMPLICATIONS OF CKD

58

2. Metabolic Acidosis

- Normal metabolism of ingested food produces approximately 1 mEq/kg of metabolic acid daily, which must be excreted by the kidneys (primarily as ammonium ion) to maintain acid-base balance
- Reduced bicarbonate reabsorption and impaired production of ammonia by the kidneys are the major factors responsible for development of metabolic acidosis in advanced kidney disease.
- Treatment: Na bicarbonate, Dialysis

COMPLICATIONS OF CKD

59

c) Anemia of CKD

- ✓ The progenitor cells of the kidney produce 90% of the hormone **erythropoietin (EPO)**, which stimulates red blood cell (RBC) production.
- ✓ Reduction in nephron mass → decreases renal production of EPO → anemia in patients with CKD.
- ✓ Anemia of CKD → decreased oxygen delivery and utilization, → **increased cardiac output** and left ventricular hypertrophy (LVH) → the cardiovascular risk and mortality in patients with CKD.
- ✓ The result is a **normochromic, normocytic anemia**.

COMPLICATIONS OF CKD

60

- ✓ Uremia decreases the lifespan of RBCs from a normal of **120** days to as low as **60** days in patients with stage 5 CKD.
- ✓ Iron deficiency and **blood loss from regular laboratory testing and hemodialysis** also contribute to the development of anemia in patients with CKD

COMPLICATIONS OF CKD

61

Treatment

- The goal of treatment for anemia of CKD is to increase Hgb levels greater than **11** g/dL

Nonpharmacologic Therapy

1. **Sufficient dietary iron intake** must be maintained in patients with anemia of CKD.
 - ✓ Approximately **1 to 2 mg of iron** is absorbed daily from the diet.
 - ✓ This small amount is generally not adequate to preserve adequate iron stores to promote RBC production.

2. RBC transfusions

COMPLICATIONS OF CKD

62

Pharmacologic Therapy

- ✓ **Iron supplementation** to correct and prevent iron deficiency caused by **ongoing blood loss** and increased iron demands associated with the initiation of erythropoietic therapy
- ✓ **ESA** to correct erythropoietin deficiency: epoetin alfa, epoetin beta and darbepoetin alfa
 - ✓ The first-line treatment for anemia of **CKD** involves replacement of erythropoietin with ESAs

COMPLICATIONS OF CKD

63

- Darbepoetin alfa differs from epoetin alfa by the addition of **carbohydrate side chains** that increase the **half-life of darbepoetin alfa** compared to epoetin alfa and endogenous EPO, allowing for **less frequent dosing** than that of epoetin alfa.
- The most common adverse effect seen with ESA is **increased blood pressure**
- The starting doses of epoetin alfa are **80** to **120** units/kg per week for **SC** administration and **120** to **180** units/kg per week for **IV** administration, divided two to three times per week.
- The starting dose of darbepoetin alfa is **0.45** mcg/kg administered **SC** or **IV** once weekly

COMPLICATIONS OF CKD

64

4. Secondary Hyperparathyroidism and Renal Osteodystrophy

- As renal function declines in patients with CKD, **decreased phosphorus excretion** disrupts the balance of calcium and phosphorus homeostasis.
- The rise in serum phosphate causes **increased binding of phosphate with calcium** in the plasma, thus **decreasing the plasma serum ionized calcium concentration**
- The parathyroid glands release PTH in response to **decreased serum ionized calcium** and **increased serum phosphorus levels**.

COMPLICATIONS OF CKD

65

- **The actions of PTH include:**
 - ✓ **Increasing calcium resorption** from bone
 - ✓ **Increasing calcium reabsorption** from the proximal tubules in the kidney
 - ✓ **Decreasing phosphorus reabsorption** in the proximal tubules in the kidney
 - ✓ **Stimulating activation of vitamin D** by 1- α -hydroxylase to calcitriol (1,25-dihydroxyvitamin D₃) to promote calcium absorption in the GI tract and increased calcium mobilization from bone

COMPLICATIONS OF CKD

66

- Calcitriol : the active form of vitamin D
 - ✓ Increases gut absorption of calcium
 - ✓ Suppress PTH release (negative feedback)
 - As kidney function continues to decline and the GFR falls less than **60**,
 - ✓ Phosphorus excretion continues to decrease (hyperphosphatemia)
 - ✓ Calcitriol production decreases
- ⇒ **Secondary hyperparathyroidism (sHPT)**

COMPLICATIONS OF CKD

67

Treatment of Secondary Hyperparathyroidism

□ Nonpharmacologic Therapy

- ✓ Dietary phosphorus restriction
- ✓ Restriction of aluminum exposure and
- ✓ Parathyroidectomy(considered in severe or refractory cases)

COMPLICATIONS OF CKD

68

Pharmacologic Therapy

i. Phosphate-Binding Agents

- ✓ Used to bind **dietary phosphate** in the GI tract to form an **insoluble complex** that is excreted in the feces
- i. Calcium-based phosphate binders:
 - ✓ **Calcium carbonate** (40% elemental Ca) and **calcium acetate** (25% elemental Ca)
 - ✓ Starting Dose **0.5–1 g** elemental Ca 3xday
 - ✓ Adverse effects: constipation, hypercalcemia, vascular calcification(Use **Sevelamer-no calcium**).

COMPLICATIONS OF CKD

69

ii. **Aluminum/magnesium-based phosphate binders:**

- ✓ Aluminum hydroxide, aluminum carbonate,
- ✓ Magnesium hydroxide, magnesium carbonate
 - Not recommended for chronic use

iii. **Lanthanum carbonate:**

- ✓ Quickly dissociates in the acidic environment of the stomach, where the lanthanum ion binds to **dietary phosphorus**, forming an insoluble compound that is excreted in the feces

2. Calcimimetic Therapy (Cinacalcet): Enhances calcium-sensing receptor sensitivity, reducing PTH secretion

COMPLICATIONS OF CKD

70

2. Vitamin D (calcitriol)

- ✓ Vitamin D occurs naturally as **ergocalciferol** (vitamin D₂) and **cholecalciferol** (vitamin D₃), both of which are inactive precursors of active forms of vitamin D.
- ✓ An intermediate activation step (25-hydroxylation) occurs in the liver to produce 25-hydroxy vitamin D (**25-hydroxycalciferol**), which is also relatively inactive.
- ✓ Final activation (1-hydroxylation) occurs in the kidney, yielding **calcitriol** (1, 25-dihydroxycholecalciferol)
- ✓ Exogenous vitamin D compounds that mimic the activity of calcitriol decrease PTH secretion (**calcitriol, paricalcitol, doxercalciferol, alfacalcidol**)
- ✓ Dosing: usually **0.25–0.5** mcg/day

Renal Replacement Therapy(RRT)

71

- Patients who progress to ESRD require RRT
- The modalities that are used for RRT are
 1. Dialysis
 2. Kidney transplantation

GLOMERULONEPHRITIS

Objectives

73

- At the end of this session, students will be able to:
 - ▣ To identify various causes of glomerulonephritis
 - ▣ Differentiate between nephrotic and nephritic syndrome
 - ▣ Recommend appropriate treatment for glomerulonephritis
 - ▣ Identify monitoring parameters for nephrotic and nephritic syndromes

Introduction

74

- The glomerulus consists of two important components:
 1. The filtration barrier (capillary wall)
 - Consists of three well-defined layers:
 - ✍ Fenestrated endothelium,
 - ✍ Glomerular basement membrane (GBM),
 - ✍ Epithelial cell layer (podocytes)
 2. The mesangium
 - Provides support for the glomerular capillaries

Introduction

76

- The passage of solutes through the glomerular membrane is impacted by both the **size** and **charge** of the solute.
- The unique capillary bed of the glomerulus;
 - ❖ Allows free passage of **small** nonprotein plasma constituents up to the size of inulin (MW of 5.2 kDa)
 - ❖ Excludes macromolecules equal to or larger than albumin (MW of 69 kDa)

Introduction

77

- Fixed, **negatively charged** sites are found within the glomeruli in all three layers of the capillary wall.
- The movement of **negatively charged molecules** is thus restricted more than that of **neutral** or **positively** charged molecules.
- Different glomerular diseases affect this size - and charge-selective barrier to different extents

Pathogenesis

78

- **Genetic mutations** producing familial disease
- **Systemic hypertension and atherosclerosis** can produce pressure stress, ischemia, or lipid oxidants that lead to chronic glomerulosclerosis
- **Diabetic nephropathy**: thickening of the GBM secondary to the long-standing effects of **hyperglycemia**, advanced **glycosylation** end products, and reactive oxygen species.
- **Infections**: Bacteria, fungi, and viruses can directly infect the kidney producing their own antigens.
- **Poststreptococcal glomerulonephritis (PSGN)**
- **Autoimmune diseases**

Clinical Presentation

79

- Patients with glomerular disease are often categorized into one of two broad classifications:
 - ⇒ **Nephritic syndrome**
 - Reflects glomerular ***inflammation*** and frequently results in ***hematuria***
 - ⇒ **Nephrotic syndrome**
 - ***Noninflammatory injury*** to the glomerular structures and results in few cells or cellular casts in the urine

Nephritic Syndrome

80

- Acute nephritic syndromes classically present with
 - ✍ Hypertension,
 - ✍ Hematuria,
 - ✍ Red blood cell casts,
 - ✍ Pyuria, and
 - ✍ Mild to moderate proteinuria(<3.5 g/day per 1.73 m²)
- Extensive inflammatory damage to glomeruli causes a fall in GFR and eventually produces uremic symptoms with salt and water retention, leading to edema and hypertension

Nephritic Syndrome

81

□ Poststreptococcal Glomerulonephritis

- Affects children between the ages of 2 and 14 years
- Skin (impetigo) and throat infections (pharyngitis) with particular M types of streptococci (nephritogenic strains) antedate glomerular disease
- PSGN due to impetigo develops 2–6 weeks after skin infection and 1–3 weeks after streptococcal pharyngitis.
- Treatment is supportive, with control of hypertension, edema, and dialysis as needed.
- Antibiotic treatment for streptococcal infection should be given to all patients and their cohabitants (active streptococcal infection or in an outbreak setting to control infection)

Nephritic Syndrome

82

- Other causes
 - ➔ Subacute Bacterial Endocarditis
 - ➔ Lupus Nephritis
 - ➔ Antiglomerular Basement Membrane Disease
 - ➔ IgA Nephropathy

Nephrotic Syndrome

83

- **Nephrotic syndrome** classically presents with
 - ➔ Heavy proteinuria (>3.5 g/day per 1.73 m²),
 - Urine dipstick 3+/4+
 - ➔ Hypoalbuminemia, <2.5 g/dL
 - ➔ Hypercholesterolemia, >200 mg/dL
 - ➔ Edema, and
 - ➔ Hypertension
 - ➔ Hypercoagulable state

Nephrotic Syndrome

84

□ Causes

- ⇒ **Minimal Change Disease:** alters capillary charge and podocyte integrity
- ⇒ **Focal Segmental Glomerulosclerosis**
- ⇒ **Membranous Glomerulonephritis:** malignancy (solid tumors of the breast, lung, colon), infection (hepatitis B, malaria, schistosomiasis), or rheumatologic disorders like lupus or rarely rheumatoid arthritis
- ⇒ **Diabetic Nephropathy**

Diagnosis

85

- **Laboratory evaluation**

- **⇒ Urinalysis**

- To determine **nephrotic nature** of glomerular disease
 - ✓ Proteinuria, $>3.5 \text{ g/day/1.73 m}^2$
 - ✓ Lipiduria
- To determine **nephritic nature** of glomerular disease
 - ✓ Hematuria
 - ✓ Pyuria
 - ✓ Cellular, granular casts

- **⇒ Glomerular filtration rate**

- To determine extent of glomerular damage

Diagnosis

86

➔ Other tests

- To identify type and etiology of glomerular disease
 - Serum complement concentration
 - Antinuclear and anti-DNA antibodies
 - Antistreptolysin antibodies: PSGN
 - Circulating antiglomerular basement membrane antibodies
 - Cryoglobulins

➔ Percutaneous renal biopsy

- To provide definitive diagnosis of glomerular disease

Treatment

87

Supportive Therapy

- **Edema:**
 - Salt restriction, bedrest, and use of support stockings
 - Diuretics:
 - Furosemide: 160 to 480 mg
- **Hypertension**
 - Goal: < 130/80 mm Hg
 - ACEI/ARBs
 - CCB (nondihydropyridines)
- **Proteinuria**
 - Dietary protein restriction
 - ACEI/ARB

Treatment

88

Supportive, continued

□ Hyperlipidemia

- A low-fat diet
- Statins: lovastatin, pravastatin, simvastatin, and fluvastatin

□ Anticoagulation

- Warfarin
 - Patients who have documented thromboembolic episodes

Treatment

89

□ Prednisolone

- ✓ **Initial:** 2 mg/kg/day or 60 mg/m²/day (maximum: 80 mg/day) in divided doses until **urine is protein free** for 3 consecutive days (maximum: 28 days); followed by 1-1.5 mg/kg/dose or 40 mg/m²/dose given every other day for 4 weeks
- ✓ **Maintenance** (for frequent relapses): 0.5-1 mg/kg/dose given every other day for 3-6 months with tapering

□ Nonsteroidal therapy

- ▣ Cyclophosphamide at a dose of 2 mg/kg per day

Treatment

90

Common definitions of patients with **nephrotic syndrome** related to **steroid treatment outcome**

- ↪ **Remission**: urine protein: creatinine ratio < 0.2 or dipstick negative or trace reading for 3 consecutive days
- ↪ **Corticosteroid responsive**: attainment of complete remission with corticosteroid therapy

Treatment

91

- ✚ **Corticosteroid resistance:** inability to induce a remission within 4 weeks of daily corticosteroid therapy
- ✚ **Relapse:** increase in first morning urine protein: creatinine ratio to ≥ 2 or dipstick reading of $\geq 2+$ for 3 of 5 consecutive days
- ✚ **Infrequent relapse:** 1–3 relapses annually
- ✚ **Frequent relapse:** ≥ 2 relapses within 6 months after initial therapy or ≥ 4 relapses in any 12-month period
- ✚ **Corticosteroid dependent:** relapse during taper or within 2 weeks of discontinuation of corticosteroid therapy

Treatment

92

- Patients who **relapse** are treated with another course of prednisone with the goal of decreasing the chance of corticosteroid toxicity by weaning soon after the patient is in remission.
 - ▣ 60 mg/m² per day of **corticosteroids** as with initial presentation with weaning to alternate-day therapy (40 mg/m² per dose) after a urine dipstick test result is negative or trace for protein for 3 consecutive days
 - Infrequent relapsers: **4-week** alternate-day therapy
 - Frequent relapse: **3-month** alternate-day therapy
- **Corticosteroid resistant MCNS**: cyclophosphamide, levamisole, cyclosporine

Evaluation of Therapeutic Outcomes

93

- For therapeutic response and treatment-related toxicities
 - ➔ Serum creatinine concentration and creatinine clearance
 - ➔ Blood pressure should be monitored periodically
 - ➔ Serum lipid concentrations
 - ➔ Urinalysis and a complete blood count
 - ➔ Patient's appetite and energy level

Thank you!